

Atiqul Islam Chowdhury

AI & ML ENGINEER · SENIOR DATA SCIENTIST

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Experience

BRAC Bank PLC

Dhaka, Bangladesh

ASSOCIATE MANAGER, R&D (PREDICTIVE AI)

Apr. 2025 - Present

- Architect and deploy end-to-end ML pipelines — from data ingestion and feature engineering to model training, evaluation, and on-premise production deployment on Windows Server infrastructure.
- Develop and maintain secure, scalable AI/ML solutions in adherence with data privacy regulations (GDPR-equivalent standards) and internal compliance frameworks.
- Designed and implemented customer segmentation and personalization models using clustering and RFM algorithms, improving targeting precision for banking products.
- Fine-tune predictive models for credit risk scoring, churn prediction, and revenue forecasting, ensuring strict compliance with banking regulations.
- Engineering end-to-end data pipelines to fetch, cleanse, and transform raw transactional data, including designing and optimizing database procedures to ensure production-grade data reliability and readiness for downstream ML and reporting use cases.
- Developed an ATM cash-loading optimization model to forecast cash demand across branches, reducing idle cash and cash-out incidents while improving cash provisioning efficiency.
- Built an occupation-based transaction portfolio monitoring model for AML compliance, profiling customer transaction behavior against declared occupation to flag anomalous patterns for investigation.
- Designed and deployed a fraud detection model for bKash transactions, leveraging behavioral and transactional features to identify suspicious activity in near real-time and reduce false positives.
- Build interactive executive-level dashboards in Power BI, integrating ML-driven KPIs to support strategic decision-making across departments.
- Mentoring junior analysts on ML best practices, model explainability, and responsible AI principles.

Next Ventures

Dhaka, Bangladesh

SENIOR DATA ANALYST

Feb. 2024 - Apr. 2025

- Conducted in-depth exploratory data analysis (EDA) and statistical modeling to surface actionable business insights, directly influencing product roadmap decisions.
- Built and deployed ML models (regression, time-series forecasting with Prophet/ARIMA) for revenue prediction, achieving measurable improvements in forecast accuracy.
- Designed and maintained interactive dashboards in Metabase and Looker Studio, enabling real-time monitoring of key product metrics and business KPIs.
- Architected and implemented database replication pipelines for MySQL and PostgreSQL using AWS Database Migration Service (DMS), ensuring high availability and zero-downtime migrations.
- Created comprehensive Data Flow Diagrams (DFDs) and technical documentation to maintain transparency across engineering and business stakeholders.

FinSource Limited

Dhaka, Bangladesh

TEAM LEAD (DATA)

Jun. 2023 - Jan. 2024

- Managed the Data team by maintaining day-to-day tasks.
- Maintained accurate records of participants in the 401(k) plans.
- Developed and optimized complex database objects - stored procedures, functions, triggers, and views — to support scalable application development.
- Developed database functions, scripts, stored procedures, and triggers to support application development.
- Worked with cross-functional teams to address data-related challenges and supported decision-making processes.
- Generated reports for plan sponsors and participants to ensure compliance with legal requirements and monitored the plan's performance.

FinSource Limited

Dhaka, Bangladesh

DATA ANALYST

Aug. 2021 - May. 2023

- Worked on US Retirement Plan Services.
- Developed Crystal Report/API for data migration, scrubbing, and mapped files to integrate with different stakeholders' systems.
- Produced and maintained operational reports providing essential business metrics across departments, supporting executive-level decision-making.
- Wrote programming scripts to enhance the functionality and performance of company applications.
- Developed database functions, scripts, stored procedures, and triggers to support application development.
- Developed, maintained, and generated reports that provided essential aspects to colleagues from all departments and provided admin support for Salesforce.

TallyKhata

Dhaka, Bangladesh

ASSOCIATE DATA ANALYST

Dec. 2020 - Jul. 2021

- Designed SQL/PL-SQL scripts and automated Python scripts for data extraction, reporting, and stakeholder email generation.
- Created investor-facing dashboards in Google Data Studio (Looker Studio) for board meetings, presenting key financial and growth metrics.
- Developed Python scripts for ETL (Extract, Transform, and Load).
- Built ML classification models to predict user behavior categories, supporting data-driven product decisions for a fintech platform.
- Built Machine Learning models to predict different user categories.

Pioneer Alpha

Dhaka, Bangladesh

RESEARCH & DATA ANALYST

Dec. 2019 - Dec. 2020

- Developed research projects and made them into publications.
- Collected, organized, and interpreted various organizational data.
- Analyzed the data to inform organizational decisions.
- Built various Machine Learning and Neural Network models for various research projects.

Education

Ahsanullah University of Science and Technology

Dhaka, Bangladesh

B.SC. IN COMPUTER SCIENCE AND ENGINEERING

Jun. 2015 - Jul. 2019

- Thesis: A Study on Image Processing to Facilitate Business System by Multiple Barcode Detection

Institute of Business Administration, University of Dhaka

Dhaka, Bangladesh

ADVANCED CERTIFICATE FOR MANAGEMENT PROFESSIONALS (ACMP) 4.0

Jul. 2025 - Sep. 2025

Honors, Awards & Community

Jul. 2024 –
Present

Data Instructor, Data Analytics Career Bootcamp — Human Development Network Bangladesh (HDNB)

Dhaka, Bangladesh

Oct, 2024 **Judge**, 1st Data Innovators Challenge by AI Expert Career

DIPTI, Dhaka,
Bangladesh

Sep, 2024 **Guest Speaker**, Exploring Data Analytics: Basic & Career Opportunities

UITS, Dhaka,
Bangladesh

Skills

Programming	C, Python, R, SQL/PL-SQL
Machine Learning	XGBoost, LightGBM, Random Forest, SVM, KNN, Logistic & Linear Regression, K-Means, Scikit-learn
Deep Learning & AI	TensorFlow, Keras, CNNs, RNNs/LSTMs, Transfer Learning, NLP, Computer Vision, Generative AI/LLMs
Data Science	Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy, OpenCV, Prophet
Database Management	Oracle, PostgreSQL, MySQL, PL/SQL
Database Migration Service	AWS Database Migration Service, DBConvert
Data Visualization Tools	Looker Studio, Metabase, Superset, Microsoft Power BI, Tableau
Reporting Tools	Crystal Reports, MS Office
Web Development	HTML, CSS, JavaScript
Task Management Tools	ClickUp, Notion, Azure DevOps

Certifications

Applied Machine Learning: Algorithms

LINKEDIN LEARNING

Dec. 2024

- Credential: <https://shorturl.at/2Bm10>

Google Advanced Data Analytics (Specialization)

COURSERA

Apr. 2026

- Credential: <https://www.coursera.org/account/accomplishments/specialization/81D99K5WIMRZ>

Google Data Analytics (Specialization)

COURSERA

Mar. 2022

- Credential: https://drive.google.com/file/d/1vT1AsIPaH2INzdElwE70_y2240uRsRoW/view

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional

COURSERA

Oct. 2025

- Credential: <https://shorturl.at/ck0jj>

SQL (Advanced) Certificate

HACKERRANK

Aug. 2022

- Credential: <https://www.hackerrank.com/certificates/3aa2e454b635>

SQL (Intermediate) Certificate

HACKERRANK

Apr. 2021

- Credential: <https://www.hackerrank.com/certificates/7d5d74f387f5>

Understanding Machine Learning

DATA CAMP

Mar. 2020

- Credential: <https://shorturl.at/H0tUt>

Publications

An Exploration of Machine Learning and Deep Learning-Based Diabetes Prediction Techniques

Springer

BOOK CHAPTER

Sep. 2022

- Authors: **AI Chowdhury**, KA Mamun
- Emerging Technologies in Data Mining and Information Security Conference paper
- Full Read: https://link.springer.com/chapter/10.1007/978-981-19-4676-9_23

A Study on Multiple Barcode Detection from an Image in Business System

IJCA

JOURNALS

Jan. 2019

- Authors: **AI Chowdhury**, MS Rahman, N Sakib
- International Journal of Computer Applications (IJCA), Volume: 181, No: 37, pp: 30-37
- Full Read: <https://www.ijcaonline.org/archives/volume181/number37/30275-2019918340>

A Survey on Dimensionality Reduction Techniques for Time-Series Data

IEEE Access

JOURNALS

Apr. 2023

- Authors: M Ashraf, F Anowar, JH. Setu, **AI Chowdhury**, E Ahmed, A Islam, A Al-Mamun
- Advances in Intelligent Systems and Computing (AISC) series
- Full Read: <https://ieeexplore.ieee.org/document/10107391>

An Automated System in ATM Booth Using Face Encoding and Emotion Recognition Process

ACM

CONFERENCE

Aug. 2020

- Authors: **AI Chowdhury**, MM Shahriar, A Islam, E Ahmed, A Karim, MR Islam
- 2nd International Conference on Image Processing and Machine Vision (IPMV)
- Full Read: <https://dl.acm.org/doi/10.1145/3421558.3421567>

hActNET: An Improved Neural Network based Method in Recognizing Human Activities

IEEE

CONFERENCE

Oct. 2020

- Authors: **AI Chowdhury**, M Ashraf, A Islam, E Ahmed, MS Jaman, MM Rahman
- 4th International Symposium on Multidisciplinary Studies and Innovative Technologies
- Full Read: <https://ieeexplore.ieee.org/document/9254992>

A Framework for Lung Cancer Detection at Early Stages with IoT and Decision Support System

IEEE

CONFERENCE

Nov. 2023

- Authors: **AI Chowdhury**, S. Jahan, TM Mahee, MI Khan, MT Habib, A Islam
- International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT)
- Full Read: <https://ieeexplore.ieee.org/document/10391757>

Acknowledgement

I hereby declare that all information provided in this curriculum vitae is accurate and complete to the best of my knowledge. I take full responsibility for the authenticity of the details presented herein.